General Mathematics — Exercise Sheet 12

Vectors, length and projection

Variant: Statements

Exercise 1 (Recovering a point from displacements). The points $P = (2, -1, 3)$ and $Q = (-1, 4, 2)$ are given in Cartesian coordinates. Let $\boldsymbol{z} = (1, 2, -2)$. Find the point R satisfying

$$\overrightarrow{QR} = 2\overrightarrow{PQ} - \boldsymbol{z}.$$

Give the standard coordinate representation of the position vector of R, and verify the displacement equation from your answer.

Exercise 2 (An orthogonality parameter). For $a \in \mathbb{R}$, let $\boldsymbol{u}(a) = (a, 2, -1)$ and $\boldsymbol{v} = (2, -1, 3)$. Find all a for which these vectors are orthogonal. At that value of a, find the unit vector in the direction of $\boldsymbol{u}(a)$. Also find the angle between $\boldsymbol{u}(1)$ and $\boldsymbol{v}$.

Exercise 3 (Prescribing two direction angles). Find every nonzero vector $\boldsymbol{u} \in \mathbb{R}^3$ whose angle with $\boldsymbol{e}_1$ is $\pi/3$ and whose angle with $\boldsymbol{e}_2$ is $\pi/4$. Give every possible angle with $\boldsymbol{e}_3$. The angle is always the value in $[0, \pi]$ from Lecture 12.

Exercise 4 (A closest point with a parameter constraint). Let $P = (2, 3, -1)$ and $A = (1, 0, 2)$. A moving point has position

$$X(t) = A + t\boldsymbol{v}, \qquad \boldsymbol{v} = (1, 2, 2), \quad t \in \mathbb{R}.$$

Find the point on the whole line closest to P, and its distance from P. Then find the closest permitted point when $1 \le t \le 2$. Explain why the unrestricted answer is not permitted in the second task.

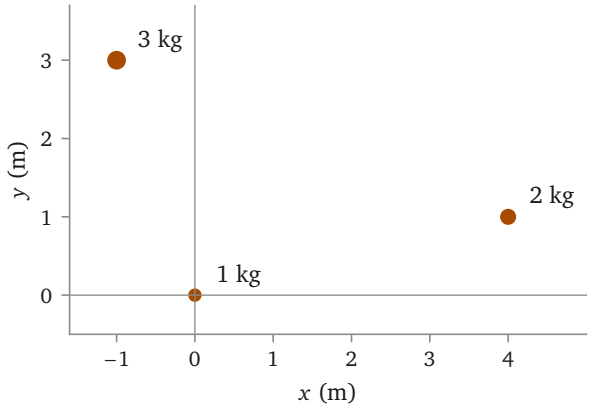
Exercise 5 (A given centre-of-mass model). Three point particles have masses $m_1 = 1$, $m_2 = 2$ and $m_3 = 3$ kilograms, and position vectors in metres

$$\boldsymbol{r}_1 = (0, 0), \qquad \boldsymbol{r}_2 = (4, 1), \qquad \boldsymbol{r}_3 = (-1, 3).$$

For positive point masses, the given model defines their *centre of mass* by

$$\boldsymbol{R} = \frac{\sum_j m_j \boldsymbol{r}_j}{\sum_j m_j}.$$

The diagram shows the given positions and masses.



1. Compute $\boldsymbol{R}$ and verify $\sum_{j=1}^{3} m_j(\boldsymbol{r}_j - \boldsymbol{R}) = \boldsymbol{0}$ for these data.
2. Translate every position by $\boldsymbol{a} = (-2, 1)$ metres. Compute the new centre and compare it with $\boldsymbol{R} + \boldsymbol{a}$.
3. Replace the first two particles by their combined mass at their centre of mass. Check that the centre of the resulting two-particle system is still $\boldsymbol{R}$.

The centre-of-mass formula is supplied as a physical model, not derived from a force law.